

HW #1: Due Friday, June 6, 10:00 am CST.

I. Do Problems 1.1, 1.3, and 1.4 from the textbook.

II. In the simple regression model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, one usually deals with pairs of observations $(x_i, y_i), i = 1, \dots, n$. Show that

1. $\sum_{i=1}^n (x_i - \bar{x}) = 0$, and $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})y_i$.

2. $S_x^{-2} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n c_i y_i$,

where $c_i = \frac{x_i - \bar{x}}{S_x^2}$, $S_x^2 = \sum_{i=1}^n (x_i - \bar{x})^2$.

NOTES: (i) The bonus problems tend to get deeper into some statistical concepts, they will encourage you to review/study the relevant sections of this or your earlier textbook/notes more systematically.

(ii) To get the bonus points for a problem, a complete solution must be given. No partial credits for the Bonus problems.

(iii) **Bonus HW points will be banked**, and used only in borderline cases when assigning letter grades at the end of the summer.

3. (Bonus) For the simple linear regression model, $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$,

(a) Derive the equations for minimizing the residual sum of squares (RSS):

$$Q(\beta_0, \beta_1) = \sum (y_i - \beta_0 - \beta_1 x_i)^2,$$

and give the formulae for the least squares estimates (LSE) $\hat{\beta}_0, \hat{\beta}_1$ of the regression coefficients.

(b) Show that $\hat{\beta}_1 = \sum_{i=1}^n c_i y_i$, with c_i 's as above. If $\varepsilon_i \sim N(0, \sigma^2)$, what is the distribution of $\hat{\beta}_1$?

(c) Let $e_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$, $i = 1, 2, \dots, n$, be the (estimated regression) residuals. Show that

$$\sum_{i=1}^n e_i = 0, \text{ and } \sum_{i=1}^n e_i x_i = 0.$$

What is the (geometrical) interpretation of the above identities?